1. Linear Combinations of Random Variables

1.1. Expectation & Variance of a Function of $\mathbf{X}$

$$E(aX+b) = aE(X)+b$$

$$\text{Var}(aX+b) = a^2\text{Var}(X)$$

{IS Ex 6a} Question 12:

The random variable T has mean 5 and variance 16. Find two pairs of values for the constants c and d such that $E(cT+d) = 100$ and $\text{Var}(cT+d) = 144$

Solution:

Expand expectation equation:

$$E(cT+d) = cE(T)+d = 100$$

$$\therefore 5c+d = 100$$

Expand variance equation:

$$\text{Var}(cT+d) = c^2\text{Var}(T) = 144$$

$$16c^2 = 144$$

$$c = \pm 3$$

Use first equation to find two pairs:

$$c = 3,\ \ d = 85c = -3,\ \ d = 115$$

1.2. Combinations of Random Variables

- Expectations of combinations of random variables:

$$E(aX+bY) = aE(X)+bE(Y)$$

- Variance of combinations of independent random variables:

$$\text{Var}(aX+bY+c) = a^2\text{Var}(X)+b^2\text{Var}(Y)$$

$$\text{Var}(X \pm Y) = Var(X)+Var(Y)$$

- Combinations of identically distributed random variables having mean μ and variance σ^2

$E(2X) = 2\mu$ and $E(X_1)+E(X_2) = 2\mu$

$\text{Var}(2X) = 4\sigma^2$ and $\text{Var}(X_1+X_2) = 2\sigma^2$

{IS Ex 6b} Question 3:

It is given that X_1 and X_2 are independent, and $E(X_1) = E(X_2) = \mu$, $\text{Var}(X_1) = Var(X_2) = \sigma^2$. Find $E(\overline{X})$ and $\text{Var}(\overline{X})$, where $\overline{X} = \frac{1}{2}(X_1+X_2)$

Solution:

Split the variance into individual components

$$\text{Var}\left(\frac{1}{2}(X_1+X_2)\right) = \left(\frac{1}{2}\right)^2\text{Var}(X_1)+\left(\frac{1}{2}\right)^2\text{Var}(X_2)$$

Substitute given values, hence

$$\text{Var}\left(\frac{1}{2}(X_1+X_2)\right) = \frac{1}{4}\sigma^2+\frac{1}{4}\sigma^2 = \frac{1}{2}\sigma^2$$

1.3. Expectation & Variance of Sample Mean

$E(\overline{X}) = \mu$, $\text{Var}(\overline{X}) = \frac{\sigma^2}{n}$

{IS Ex 6c} Question 5:

The mean weight of a soldier may be taken to be 90kg, and $\sigma = 10$ kg. 250 soldiers are on board an aircraft, find the expectation and variance of their weight. Hence find the μ and σ of the total weight of soldiers.

Solution:

Let X be the average weight, therefore

$$E(\overline{X}) = \mu = 90$$

$\text{Var}(\overline{X}) = \frac{\sigma^2}{n} = \frac{10^2}{250} = 0.4$ kg^2

To find μ of total weight, you are calculating

$E(X_1)+E(X_2)\ldots+E(X_{250}) = 250E(X) = 22\,500$ kg

To find σ, first find $Var(X)$

$\text{Var}(X_1)\ldots+Var(X_{250}) = 250Var(X) = 2500$ kg

$$\text{Var}(X) = \sigma^2 = 25000$$

$\therefore \sigma = \sqrt{25000} = 158.1$ kg

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